

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$$

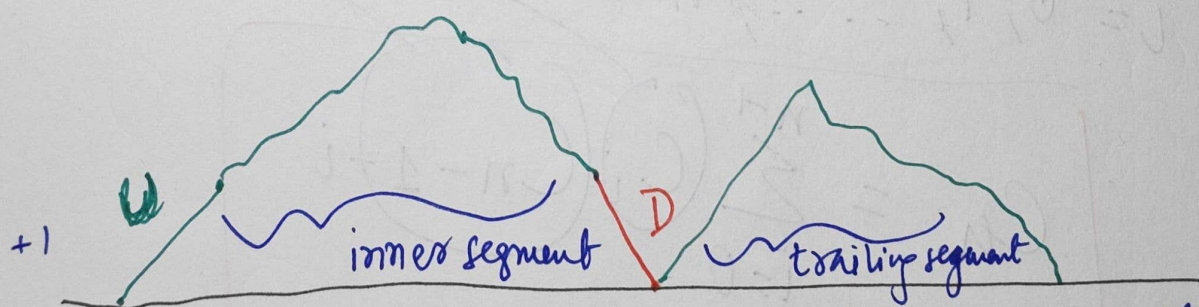
$C \rightarrow$ Catalan numbers.

$C_0 = 1$. (empty path).

Observation.

Starting slash : 'U'. opening. (else the path immediately become bad).

Point where it reaches the sea level for the first time $\rightarrow$ closing 'D'.



inner segment $\rightarrow$ subtract 1 from the height & it is a Dyck path in its own right.

$\hookrightarrow$ U's & D's are balanced.
even length $2i$

$\hookrightarrow C_i$ ways.

Trailing segment: Another Dyck

$$2n - 2i - 2$$

opening U & closing D.

$$= 2(n - i - 1). \leftarrow \text{length of the trailing segment}$$

No. of ways: C_{n-1-i}

$$(1 \text{ way to place opening U}) \times C_i \times (1 \text{ way to place closing D}) \times C_{n-1-i}$$

$$= C_i C_{n-1-i}$$

$$i = 0, 1, \dots, n-1$$

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$$

Sanity Check.

$$C_0 = 1$$

$$C_1 = C_0 C_0 = 1$$

$$C_2 = C_0 C_1 + C_1 C_0 = 1 + 1 = 2.$$

$$C_3 = C_0 C_2 + C_1 C_1 + C_2 C_0 = 2 + 1 + 2 = 5$$

$$= C_0(C_0 C_1 + C_1 C_0) + C_1 C_1 + (C_0 C_1 + C_1 C_0) C_0$$

LOGIC.

This week
Propositional
logic

Next week
Predicate logic.

Assertion

→ sentences.

↓
propositions / statements

↓
truth value.

True (1)

False (0).

A proposition can be either true or false but not both.

Logic dealing with propositions - propositional logic.

precise
truth
value.

p : Combinatorics is a required course for sophomores.

q : Margaret Mitchell wrote "Gone with the Wind"

r : $2+3=5$

What a beautiful evening! | $\rightarrow$ No truth value possible.
Get up & have breakfast

$p, q, r \rightarrow$ primitive statements $\rightarrow$
no way to break them any further
to simpler propositions.

$\neg p$ ("not p ") $\rightarrow$ negation.

$\hookrightarrow$ Combinatorics is not a reqd
subject for sophomores.

Combining propositions.

① Conjunction: $p \wedge q \longrightarrow p \text{ 'AND' } q$

② Disjunction: $p \vee q \longrightarrow p \text{ 'OR' } q$.

is true if $\vee \longrightarrow$ inclusive OR.

either p is true
or
 q is true

or both p and q are true.

$p \underline{\vee} q \longrightarrow p \text{ 'exclusive OR' } q$.

is true if

either p is true
or
 q is true

but not both are true.

(3)

$$p \rightarrow q$$

p implies q .

p is sufficient for q .

$\hookrightarrow$ sufficient condition

p only if q .

q is necessary for p

$\hookrightarrow$ necessary condition.

p (hypothesis), q (conclusion).

if $\rightarrow$ then.

If combinatorics is a required course for sophomores then Margaret Mitchell wrote "Gone with the Wind"

④ Biconditional : $p \leftrightarrow q$.

"p if and only if q"

"p is necessary & sufficient for q"

"p iff q"

The number x is an integer

x is a variable

Truth values of propositions & their different combinations $\rightarrow$ summarized

TRUTH TABLE

p	$\neg p$
0	1
1	0

p	q	$p \wedge q$	$p \vee q$	$p \vee \neg q$	$p \rightarrow q$	$p \leftrightarrow q$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	1	1	0	1	1

E.g. 1.

S: Albest goes out for walk.

t: The moon is out

u: It is snowing.

$$(t \wedge \neg u) \rightarrow S$$

If (the moon is out and it is not snowing)
then Albest goes out for a walk.

$$\neg (S \leftrightarrow (u \vee t)):$$

It is not the case that (Albest goes out
for a walk if and only if it is snowing
or the moon is out).

"If it is snowing and the moon is not out, then
Albest will not go out for a walk."

$$(u \wedge \neg t) \rightarrow \neg S$$

E.g. 2.

Draw the TT for the compound statement

"Margaret Mitchell wrote Gone with the Wind

and (if $2+3 \neq 5$ then combinatorics is a
required course for sophomores)"

$$q \wedge (\neg r \rightarrow p).$$